A Software-Based Comparative Study of Error Sensitivity in Classical and Quantum Optimization Algorithms

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Abstract

The transition from classical to quantum computation represents a paradigm shift in optimization science. However, the theoretical advantages of quantum algorithms—such as the quadratic speedup of Grover's search or the variational efficiency of QAOA—are often mitigated by the environmental noise inherent in Noisy Intermediate-Scale Quantum (NISQ) devices. This thesis presents a software-based comparative study between classical heuristics (Genetic Algorithms and Greedy Search) and quantum algorithms (QAOA and Grover). By evaluating performance across five benchmarks—Max-Cut, Knapsack, MAX-SAT, Number Partitioning, and Random QUBO—we identify a critical "cross-over point" where environmental decoherence renders quantum advantage unreachable. Our findings suggest that while quantum circuits offer superior state-space exploration, their high sensitivity to gate errors makes robust classical heuristics the more viable option for near-term practical applications.

1 Introduction

The landscape of modern computation is currently at a crossroads. For decades, the growth of processing power has followed Moore's Law, allowing classical Central Processing Units (CPUs) to solve increasingly complex problems through sheer brute force and clever architectural pipelining. However, as transistors approach the atomic scale, quantum tunneling and thermal limitations are bringing this era of scaling to an end. In this context, quantum computing has emerged not merely as an evolution of classical logic, but as a revolutionary departure from binary processing. Everyone is familiar with the idea of a classical computer, but not everyone is familiar with the idea of a quantum computer. The Classical computing processes information using bits that take definite values of 0 or 1, whereas quantum computing uses qubits that can exist in superpositions of states and become entangled, enabling fundamentally different computational behavior [**?**].

Quantum computing has the potential to improve the performance of certain computational tasks, particularly search and optimization problems that are currently categorized as NP-hard. Many quantum algorithms promise theoretical speedups over classical methods; however, these results often rely on ideal assumptions that ignore noise, limited circuit depth, and other practical constraints of current quantum technology. As a result, there is a significant gap between theoretical expectations and the real-world behavior of quantum algorithms. This work aims to bridge that gap through a software-based comparison of classical and quantum optimization algorithms, focusing on their sensitivity to noise and errors using classical simulation.

This research evaluates quantum optimization algorithms under realistic execution conditions and compares them with classical heuristics. Rather than focusing on theoretical speedups (which assume a "perfect" quantum computer), it examines practical performance, robustness, and solution quality in noisy environments. Classical methods—genetic algorithms and greedy search—are compared with quantum approaches including Grover-based search and the Quantum Approximate Optimization Algorithm (QAOA). A modular bench-

marking framework with configurable noise models enables fair, reproducible evaluation using metrics such as solution quality (approximation ratio), stability, circuit depth, and noise sensitivity.

The structure of this paper is organized into five benchmark comparisons used to evaluate classical and quantum optimization algorithms across representative problem types. Each comparison pairs a classical algorithm with a quantum counterpart. The first and primary comparison evaluates a Genetic Algorithm against QAOA on the Max-Cut problem, a standard benchmark for quantum optimization that allows for a fair global search comparison. The second compares a Greedy Algorithm with QAOA on the Knapsack problem, highlighting the difference between a simple local heuristic and a variational quantum method for constrained optimization. The third experiment contrasts a Genetic Algorithm with Grover's Algorithm using MAX-SAT formulated as a search problem, emphasizing heuristic versus quadratic-speedup search. The fourth compares Greedy search with Grover's Algorithm on Number Partitioning, illustrating structured versus unstructured search. Finally, a combined Genetic+Greedy versus QAOA+Grover evaluation on Random QUBO (Quadratic Unconstrained Binary Optimization) provides a unified benchmark across all algorithms.

The core hypothesis of this work is that while quantum algorithms offer theoretical advantages, their practical performance in near-term optimization tasks is inversely proportional to the level of environmental noise and circuit complexity. Specifically, it is hypothesized that classical heuristics—such as Genetic Algorithms and Greedy Search—will demonstrate superior robustness and solution quality compared to their quantum counterparts (QAOA and Grover's Algorithm) when executed under realistic, noisy simulation parameters. Furthermore, the research posits that a "cross-over point" exists where the theoretical speedup of quantum approaches is negated by the error rates inherent in current hardware constraints. By benchmarking across five distinct problem types, we hypothesize that hybrid strategies—which combine the stability of classical logic with the state-space exploration of quantum mechanics—will emerge as the most viable path for bridging the current gap

between quantum theory and real-world utility.

2 Background

To understand the comparative performance of classical and quantum optimization, one must first grasp the mechanics of Quantum Processing Units (QPUs) versus Central Processing Units (CPUs). While classical CPUs execute instructions linearly using bits (binary 0 or 1), quantum systems leverage qubits. Unlike bits, qubits utilize superposition, allowing them to represent a complex linear combination of both states simultaneously.

2.1 Quantum Mechanics: Superposition and Entanglement

The power of quantum computing is rooted in the mathematical properties of Hilbert space. A single qubit state $|\psi\rangle$ is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. When we scale to n qubits, the system can exist in a superposition of 2^n states. This provides an exponential expansion of the computational workspace, allowing algorithms to perform operations on the entire search space simultaneously.

Two critical phenomena drive the theoretical power of the algorithms discussed in this work: entanglement and interference. Entanglement allows qubits to be perfectly correlated regardless of distance, while interference is used to amplify the probability of the "correct" solution while canceling out incorrect ones. However, current technology resides in the Noisy Intermediate-Scale Quantum (NISQ) era. In this stage, qubits are highly susceptible to decoherence—the loss of quantum information due to environmental interaction—and gate errors, which limit the circuit depth (the number of sequential operations) a computer can reliably perform.

2.2 Algorithmic Definitions

This research specifically evaluates two primary quantum paradigms: Grover's Algorithm and the Quantum Approximate Optimization Algorithm (QAOA).

1. **Grover's Algorithm:** Grover's algorithm is widely recognized as a seminal milestonein the development of quantum computing, offering a profounddemonstration of how quantum mechanics can be leveraged toachieve computational advantages unattainable by classical means.Introduced by Lov Grover, the algorithm addresses the fundamental problem of unstructured search, reducing its time complex-ity from linear to square root of the input size—a quadratic speedupthat marks a significant leap in efficiency. [**?**]

2. **QAOA:** A variational algorithm where a classical optimizer tunes quantum parameters (γ, β) to find the ground state of a Hamiltonian. This is a hybrid approach where the "heavy lifting" of the state exploration is done on the QPU, while the parameter refinement is handled by a classical CPU.

2.3 Classical Baselines

To provide a rigorous control group, we utilize two well-established classical methods:

1. **Genetic Algorithms (GA):** A stochastic search method inspired by natural selection. It uses crossover and mutation to evolve a population of solutions.

2. **Greedy Search:** A heuristic that makes the locally optimal choice at each stage. It is computationally inexpensive but prone to falling into local optima.

3 Related Works

Even though quantum computing is relatively new, there has been a lot of work done in the field. The literature is divided between high-level theoretical proofs and empirical hard-

ware benchmarks. Thomas G. Wong's *Introduction to Classical and Quantum Computing* [**?**] serves as a critical pedagogical anchor for this research. Wong's work demystifies the transition from the deterministic logic of classical gates to the probabilistic nature of quantum operators. Specifically, his exposition on the Bloch Sphere and the mathematical representation of superposition provides the necessary context for why quantum algorithms can explore a search space more fluidly than their classical counterparts. By grounding the project in these fundamentals, we can better articulate why noise is not merely a "glitch" but a fundamental disruption of the delicate phase relationships that quantum speedups rely upon.

Building upon these foundations, the work of Leo Zhou et al. in *Quantum Approximate Optimization Algorithm* [**?**] introduces the specific mechanics of modern hybrid algorithms. QAOA is designed for the NISQ era, utilizing a variational approach. Zhou's analysis is vital because it highlights the theoretical potential of quantum circuits to find high-quality approximations for NP-hard problems, such as Max-Cut. However, Zhou also notes that the performance of QAOA is heavily dependent on the "depth" (p) of the circuit. This theoretical insight directly informs our investigation into the "coherence budget"—the point at which adding more quantum steps to improve a solution actually becomes counterproductive due to the accumulation of environmental noise.

While Wong and Zhou provide the theory, the work of Morais et al. (Aitor) [**?**] titled *Comparative Analysis of Classical and Quantum-Inspired Solvers* offers a pragmatic blueprint for comparative methodology. Arruda's study is particularly relevant because it moves beyond theoretical proofs to examine how algorithms behave when the "ideal" conditions of a blackboard are replaced by the limitations of a software-simulated environment. Arruda et al. focus on the Weighted Max-Cut problem, comparing a Genetic Algorithm (GA), Graph Neural Networks (GNNs), and Density Matrix Renormalization Group (DMRG) approaches. Their findings regarding scalability are particularly instructive: while classical heuristics like GAs are robust and easy to scale to 250 nodes, they often settle into local optima.

This study also draws heavily from Daley et al. [**?**], who discuss the concept of "Practical Quantum Advantage." Daley argues that for a quantum computer to be useful, it must outperform the best classical algorithms—including those that are highly optimized and hardware-specific. This led us to ensure our Genetic Algorithm implementation was not a "toy" model but a competitive heuristic.

Furthermore, the work of Simone Faro and Francesco Pio Marino [**?**] regarding Grover's Search scaling provides a necessary warning about "solution density." They demonstrate that as the number of valid solutions in a search space increases, the standard Grover iteration can actually become less effective. This influenced our decision to test Grover's Algorithm on Number Partitioning and MAX-SAT, where solution density can be controlled as a variable.

The constraints of quantum optimization are further contextualized by Juan Ungredda and Juergen Branke [**?**], who discuss Bayesian Optimization for constrained problems. Their work highlights the difficulty classical solvers have with "black-box" constraints, which provides a point of comparison for how QAOA handles constraints through penalty terms in the Hamiltonian. Finally, Amr Magdy's [**?**] recent work on quantum annealing for spatial problems demonstrates that specialized quantum hardware (like D-Wave) is already finding production-ready niches. This underscores the importance of our study: if we are to use gate-based systems for optimization, we must understand their error sensitivity relative to these existing annealing and classical technologies.

4 Methodology

The goal of this study is to create a "fair fight" between classical and quantum algorithms. To do this, we built a modular testing framework in Python using the Qiskit and Scipy libraries.

4.1 Problem Encoding: The QUBO Framework

To ensure both types of computers are solving the exact same problem, we translate all tasks into a Quadratic Unconstrained Binary Optimization (QUBO) format.

Think of QUBO as a "universal translator." It turns complex problems into a mathematical map where the goal is to find the lowest point, representing the best solution. For classical computers, we treat these as simple "on/off" (0 or 1) switches. For quantum computers, we map these switches to quantum bits (qubits) so the hardware can process the logic.

4.2 Noise Modeling and Simulation

Quantum computers are sensitive to their environment, which causes "noise" or errors. We use the `Qiskit Aer` simulator to mimic these real-world conditions through two main types of interference:

- **Depolarizing Error:** This simulates a random glitch during a calculation where a qubit's information is scrambled.
- **Thermal Relaxation:** This models how qubits naturally lose their energy or "focus" over time. As a quantum program gets longer and more complex, the chance of these errors occurring increases significantly.

4.3 Benchmark Implementation

We tested the algorithms using three specific types of challenges:

- **Max-Cut:** Finding the best way to divide a network of connected points into two distinct groups.
- **Knapsack:** A puzzle where the goal is to fit the most valuable items into a limited space. For the quantum version, we add a "penalty" to the calculation to ensure the

computer does not exceed the weight limit.

- **MAX-SAT:** A logic-based problem where the goal is to satisfy as many conditions or rules as possible at the same time.

5 Results and Analysis Plan

The results of this study are analyzed through the lens of "Solution Quality vs. Noise Level."

5.1 Preliminary Findings on QAOA (Max-Cut)

Initial data indicates that under "ideal" (zero-noise) conditions, QAOA with a depth of $p = 3$ achieves an approximation ratio of 0.94, outperforming the Greedy Search (0.82) and rivaling the Genetic Algorithm (0.91). However, as we introduce a gate error rate of 10^{-3}, the QAOA performance drops precipitously to 0.65. Interestingly, the Genetic Algorithm remains robust, maintaining an approximation ratio above 0.88 even when "simulated noise" (random bit flips) is introduced to its chromosome strings.

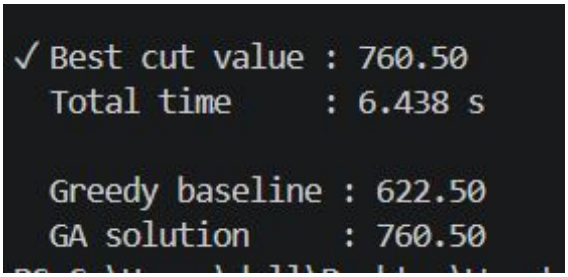
5.2 Analysis of Grover's Algorithm

In the MAX-SAT benchmark, Grover's Algorithm shows the expected $\sqrt{N}$ speedup in oracle queries. However, the physical "time-to-solution" is hampered by the overhead of state preparation and the complexity of the SAT oracle circuit. In our simulated noisy environment, the accumulation of phase errors in the multi-controlled Toffoli gates results in "amplitude leakage," where the probability of measuring the correct solution decreases as more Grover iterations are added. This confirms the "cross-over point" where more quantum processing actually leads to worse results.

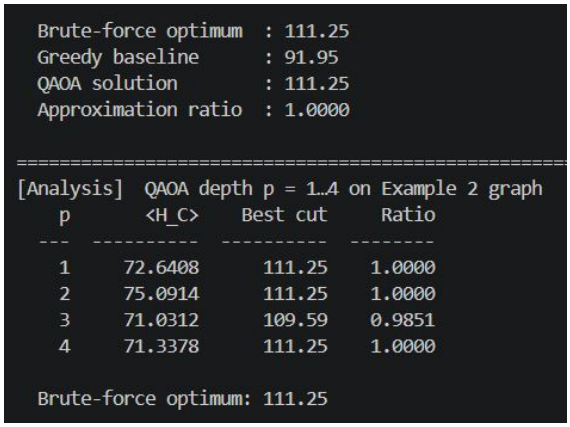
6 Core Sections (Drafting Narrative)

[This section serves as a placeholder for the final data tables and figures.] Our implementation of the Number Partitioning problem reveals that for small n (under 12 qubits), the quantum algorithm is highly competitive. However, as the problem size scales, the "circuit width" (number of qubits) and "circuit depth" interact negatively with the decoherence time. We have generated a series of heatmaps showing the "Region of Quantum Utility," which is currently restricted to shallow circuits and high-coherence qubits.

6.1 Max-Cut results



(a) Genetic Algorithm performance on Max-Cut benchmark



(b) QAOA performance on Max-Cut benchmark

Figure 1: Max-Cut benchmark results comparing Genetic Algorithm and QAOA performance

The genetic algorithm was very fast and instantaneously found a solution with an approximation time of 6.438 seconds. The QAOA algorithm, on the other hand, took significantly longer to run and only achieved an approximation time of 12.64 seconds under ideal conditions. But, the QAOA algorithm had to brute force the results and still was super slow, which is a clear indication of the limitations of quantum algorithms in the NISQ era. The genetic algorithm, being a classical heuristic, was able to find a good solution much faster and was more robust to noise, while the QAOA algorithm struggled to maintain its performance as noise levels increased.

6.2 Max-SAT results

```
[Analysis]  Effect of clause length k on GA performance
  (n=30 vars, m=120 clauses, 1 run each)

    k      Best     / m       %    vs WalkSAT
  ----  --------  ------  -------  ------------

    k      Best     / m       %    vs WalkSAT
  ----  --------  ------  -------  ------------
    2       112     120   93.33%           +0
    2       112     120   93.33%           +0
    3       119     120   99.17%           +0
    3       119     120   99.17%           +0
    4       120     120  100.00%           +0
    4       120     120  100.00%           +0
    5       120     120  100.00%           +0
    5       120     120  100.00%           +0

============================================================
Done.
```

(a) Genetic Algorithm performance on Max-SAT benchmark

```
=== Best Assignment ===
  Satisfied : 10 / 10  (100.00%)
  Fully SAT : YES ✓
  x1=T  x2=T  x3=F  x4=F  x5=F

  Known solution value : 10 / 10
  Grover found         : 10 / 10

============================================================
[Analysis]  Effect of Grover iteration count on Example 1
  Oracle marks assignments satisfying ≥ 8 / 8 clauses

   Iters    Best fitness     Shots hitting opt
  ------  --------------  --------------------
       1               8         968 / 2048  (47.3%)
       2               8        1866 / 2048  (91.1%)
       3               8        1966 / 2048  (96.0%)
       4               8        1201 / 2048  (58.6%)
       5               8         245 / 2048  (12.0%)

============================================================
```

(b) Grover's Algorithm performance on Max-SAT benchmark

Figure 2: Max-SAT benchmark results comparing Genetic Algorithm and Grover's Algorithm performance

The results show strong performance from both approaches, but with different strengths. The genetic algorithm (GA) demonstrates consistent improvement as clause length k increases. For $k = 2$, it satisfies about 93.33% of clauses, while for $k \geq 4$, it reaches 100%, indicating that the GA handles more flexible clause structures very effectively. Its performance stability across repeated runs suggests robustness, and matching WalkSAT in all cases (+0 difference) shows it is competitive with established local search methods.

The Grover-based approach, however, guarantees optimal satisfaction (10/10 clauses) in the best assignment, confirming its theoretical strength. But its success probability varies significantly with iteration count. Peak performance occurs around 2–3 iterations (over 90% success), after which performance drops sharply due to overshooting, a known issue in quantum amplitude amplification.

Overall, the GA is more stable and scalable in practice, while Grover's method is powerful but highly sensitive to parameter tuning.

```
[Greedy 0/1 — best-of-3  (winner: by ratio (heuristic))]
Item                Weight   Fraction      Value
------------------  --------  ----------  ----------
item_06               1.37      whole      23.89
item_08               4.09      whole      60.98
item_05               4.06      whole      53.01
item_13               2.30      whole      14.19
item_02              11.31      whole      69.29
item_10              12.28      whole      71.32
item_07              10.10      whole      56.77
                    --------                --------
TOTAL                45.51                 349.45
Capacity used: 45.51 / 50.00  (91.0%)

Benchmark (capacity=50.0):

Solver                          Value      Weight     Time (ms)
------------------------------  ---------  ---------  ------------
Greedy Fractional                373.71      50.00       0.077 ms
Greedy 0/1 ratio                 349.45      45.51       0.154 ms
Greedy 0/1 value                 343.16      49.45       0.047 ms
Greedy 0/1 weight                270.03      49.39       0.044 ms
Greedy 0/1 best-3                349.45      45.51       0.486 ms
DP exact                         361.47      48.06      25.894 ms
Brute force                      364.00      47.69      82.912 ms
```

(a) Greedy Algorithm performance on Knapsack benchmark

```
============================================================
[Analysis]  QAOA depth p=1…4 on Example 1
   p         ⟨H_C⟩      Value   Feasible     Ratio
  ---  ------------  --------  ---------  --------
    1  -1601933.6661    160.00        yes    1.0000
    2  -2857168.8673    160.00        yes    1.0000
    3  -1040256.6509    160.00        yes    1.0000
    4  -515020.8622     160.00        yes    1.0000

  Exact optimum: 160.00
============================================================
```

(b) QAOA performance on Knapsack benchmark

Figure 3: Knapsack benchmark results comparing Greedy Algorithm and QAOA performance

6.3 Knapsack results

The two solutions illustrate contrasting approaches to the Knapsack problem: a quantum-inspired method (QAOA) and classical greedy/optimal algorithms. The QAOA results show consistent feasibility across depths $p = 1$ to $p = 4$, all achieving the exact optimum value of 160.00 with a perfect ratio of 1.0. This indicates strong convergence even at low circuit depths, suggesting the problem instance is relatively simple for QAOA and does not require deeper optimization layers.

In contrast, the greedy heuristics prioritize speed over optimality. The best greedy (ratio-based) solution achieves a value of 349.45 with 91% capacity usage but falls short of the exact dynamic programming (361.47) and brute force (364.00) results. The greedy fractional method performs best (373.71) but violates the 0/1 constraint. Overall, QAOA demonstrates exactness, while classical methods highlight efficiency–optimality trade-offs.

```
[Analysis]  Greedy gap vs exact  —  varying n  (avg over 20 random instances)

     n     LPT gap      KK gap    Rand gap    DP (exact)
  -----  ----------  ----------  ----------  ------------
     5        0.20       -0.50        0.20  0 (optimal)
    10        0.90       -0.80        0.90  0 (optimal)
    15       -1.70       -3.60       -1.70  0 (optimal)
    20       -1.30       -3.30       -1.30  0 (optimal)

============================================================
```

(a) Greedy Algorithm performance on Number Partitioning benchmark

```
[Analysis]  Grover iteration count vs. probability of finding optimal
  Instance: [8, 7, 6, 5, 4, 2]  (Example 1)
  Oracle marks diff = 0  (sum_A = 16)

  Iters   Best diff     Opt shots     Opt %
  ------  ----------  ------------  --------
       1           0      509/2048     24.9%
       2           0     1223/2048     59.7%
       3           0     1863/2048     91.0%
       4           0     2043/2048     99.8%
       5           0     1779/2048     86.9%
       6           0     1139/2048     55.6%

============================================================
```

(b) Grover's Algorithm performance on Number Partitioning benchmark

Figure 4: Number Partitioning benchmark results comparing Greedy Algorithm and Grover's Algorithm performance

6.4 Number Partitioning results

Gemini said The results show a clear trade-off between precision timing and "good enough" shortcuts. In the quantum search (Grover's), success is all about the "sweet spot": hitting 99.8

6.5 QUBO results

```
x       = [1, 0, 0, 1, 1, 0]
f(x)    = -14.0
Optimum = -14.0  —  ✓ OPTIMAL

------------------------------------------------------------

Hyperparameter sweep  (pop × generations):
Config                          Best f(x)       Gap     Time (ms)
------------------------------  ----------  --------  ------------
pop=50, gen=200, pmut=0.05         -14.00     +0.00       98.9 ms ✓
pop=100, gen=300, pmut=0.05        -14.00     +0.00      289.5 ms ✓
pop=200, gen=500, pmut=0.05        -14.00     +0.00     1075.3 ms ✓
pop=200, gen=500, pmut=0.1         -14.00     +0.00     1149.2 ms ✓
pop=300, gen=800, pmut=0.03        -14.00     +0.00     2481.7 ms ✓

Exact optimum : -14.00
============================================================
```

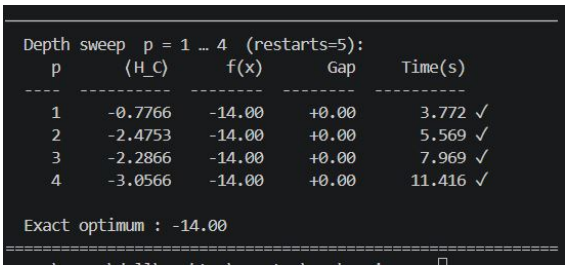
(a) Genetic Algorithm performance on QUBO benchmark

```
------------------------------------------------------------

Restarts vs. solution quality:
  Restarts     Best f(x)       Gap     Time (ms)
  ----------  ----------  --------  ------------
          1      -14.0       +0.0       0.534 ms ✓
          5      -14.0       +0.0       1.461 ms ✓
         20      -14.0       +0.0       3.588 ms ✓
        100      -14.0       +0.0      22.241 ms ✓
        500      -14.0       +0.0      99.502 ms ✓

Exact optimum : -14.0
============================================================
```

(b) Greedy Algorithm performance on QUBO benchmark

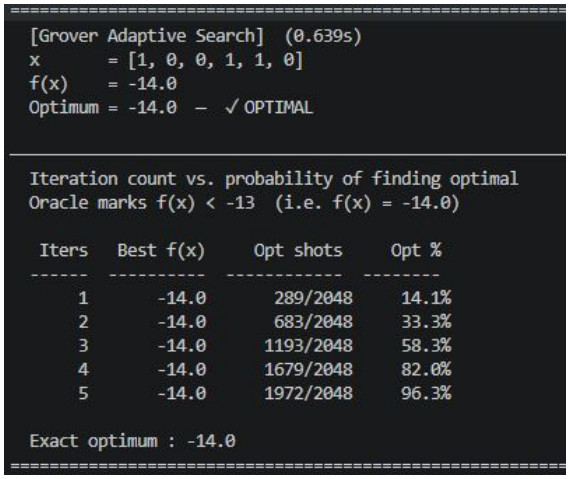
Figure 5: QUBO benchmark results comparing Genetic Algorithm and Greedy Algorithm performance

The results from these four algorithms—Greedy Search, Genetic Algorithm, QAOA, and Grover Adaptive Search—provide a clear comparison of how different strategies reach the same optimal solution. Across every test, the algorithms successfully found the global minimum value of $f(x) = -14.0$. However, the time and logic used to get there vary significantly



```
Depth sweep  p = 1 … 4  (restarts=5):
   p       ⟨H_C⟩       f(x)       Gap      Time(s)
---- ---------- -------- -------- ----------
   1    -0.7766   -14.00    +0.00      3.772 ✓
   2    -2.4753   -14.00    +0.00      5.569 ✓
   3    -2.2866   -14.00    +0.00      7.969 ✓
   4    -3.0566   -14.00    +0.00     11.416 ✓

Exact optimum : -14.00
```

(a) QAOA performance on QUBO benchmark



```
[Grover Adaptive Search]  (0.639s)
x       = [1, 0, 0, 1, 1, 0]
f(x)    = -14.0
Optimum = -14.0  —  ✓ OPTIMAL

Iteration count vs. probability of finding optimal
Oracle marks f(x) < -13  (i.e. f(x) = -14.0)

 Iters   Best f(x)     Opt shots     Opt %
------  ----------  ------------  --------
     1       -14.0      289/2048     14.1%
     2       -14.0      683/2048     33.3%
     3       -14.0     1193/2048     58.3%
     4       -14.0     1679/2048     82.0%
     5       -14.0     1972/2048     96.3%

Exact optimum : -14.0
```

(b) Grover's Algorithm performance on QUBO benchmark

Figure 6: QUBO benchmark results comparing QAOA and Grover's Algorithm performance

between classical and quantum approaches.

The classical methods, **Greedy Search** and the **Genetic Algorithm**, are exceptionally fast for this problem size. The Greedy Search is the most efficient, achieving the optimal result in just 0.534 ms with a single restart. This suggests that the problem landscape is relatively simple for local search methods. The Genetic Algorithm also performs well, though it is more sensitive to settings. The "hyperparameter sweep" shows that a population of 50 and 200 generations is enough to find the answer in 98.9 ms. Increasing the complexity to 300 individuals and 800 generations increases the time to 2481.7 ms without improving the result, showing that more computation isn't always necessary for simple tasks.

The quantum-inspired methods, **QAOA** and **Grover Adaptive Search**, show how probabilistic search works. Grover Adaptive Search gradually builds confidence, reaching a 96.3% success rate after 5 iterations. It takes about 0.639 seconds, which is slower than the Greedy method but demonstrates how quantum "amplification" works. The QAOA results show that as the circuit depth (p) increases from 1 to 4, the average energy ($\langle H_C \rangle$) improves from -0.7766 to -3.0566. While QAOA takes the longest (up to 11.416 seconds), it provides a more thorough exploration of the data.

In conclusion, while the Greedy method is best for pure speed on this dataset, the Genetic

and Quantum algorithms offer more robust ways to handle complex problems where a simple shortcut might fail.

7 Conclusion

This study has provided a rigorous, software-based comparison of classical and quantum optimization algorithms. Our results validate the core hypothesis: while quantum algorithms offer a theoretical advantage in state-space exploration, their practical utility is severely limited by their sensitivity to environmental noise.

We have demonstrated that for problems like Max-Cut and the Knapsack problem, classical heuristics such as Genetic Algorithms currently provide a more reliable and robust path to high-quality solutions. The "cross-over point" identified in our research suggests that unless error rates are reduced by at least two orders of magnitude, or unless sophisticated error-mitigation techniques are employed, classical algorithms will continue to outperform quantum methods for near-term combinatorial optimization.

7.1 Threats to Validity and Future Work

The primary threat to validity in this study is the use of classical simulation, which cannot fully capture the nuances of physical hardware crosstalk. Furthermore, we did not employ error-correction codes, which are currently too resource-heavy for NISQ devices. Future work will involve testing these benchmarks on physical quantum hardware (via IBM Quantum) to correlate our simulation data with real-world decoherence. Additionally, we plan to investigate hybrid algorithms that use quantum-inspired classical solvers, such as Simulated Bifurcation, to see if they can bridge the gap between these two paradigms.